

# Stan Serebriakov

*Software Engineer*

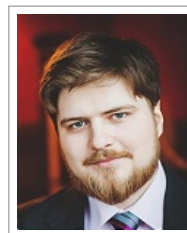
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Linkedin, StackOverflow, GitHub



## Education

- 2004–2010 **MSc in Robotics**, *St. Petersburg Polytechnic University*.  
Simultaneous Localization and Mapping with a ToF Camera
- 2010–2012 **PhD Student in Applied Maths**, *St. Petersburg Polytechnic University*.  
Monocular 3D Tracking

## Experience

- 2024–2026 **C++ Engineer**, *MPK Soft*, St. Petersburg.
- Designed and implemented a data processing server and client SDK.
  - Implemented LoRa chirp recognition and frequency-hopping table estimation.
  - Built a control system and simulator for an interceptor drone.
- Stack: C++20, Qt, Boost.Asio, FlatBuffers, FFTW, MAVLink, OpenCV, Gazebo
- 2021–2023 **iOS Engineer**, *Constanta*, Moscow.
- Implemented core features and frameworks, refactored legacy code, and reviewed merge requests for a leading sports analytics [app](#).
  - Migrated an Objective-C AR app prototype into a fully published Swift [app](#).
- 2015–2021 **Lead iOS Engineer**, *Luxoft*, St. Petersburg.
- Developed multiple iOS applications for various clients, including:
- iStockTrack**: A stock market data browser supporting voice and text messaging for communication with market [advisors](#).
  - Citibank**: A private banking app based on Apache Cordova.
  - Gazprom Neft**: A [refueling app](#).
  - Gosuslugi**: [Citizen feedback](#) and executor applications.
- Additionally, conducted hundreds of technical interviews worldwide.
- 2015–2015 **iOS Engineer**, *GuessMe*, St. Petersburg.
- Developed a social networking iOS application, allowing users to create interactive posts and share them with relevant contacts. Key contributions included:
- Developed a Swift JSON mapper [generator](#).
  - Implemented core iOS system features (networking, data management, etc.).
- 2014–2014 **Lead R&D Engineer**, *x-Turion*, St. Petersburg.
- Led software development for a compact home robot equipped with a surveillance camera for remote audio and video monitoring of private premises. Key device features included autonomous navigation, object and face recognition, and mobile app control.
- Stack: C++, ROS, OpenCV

- 2012–2014 **Mobile Engineer**, *i-Free*, St. Petersburg.  
Developed multiple iOS and Android applications, including:
- A museum events browser with ticket booking capabilities.
  - A natural language voice assistant for internet search queries.
- 2009–2011 **Lead Engineer, Co-founder**, *Cybrox*, St. Petersburg.  
Led the development of 3D IR-based head trackers and conducted research on webcam-based 6D head tracking.
- Implemented an OpenCV-based prototype and ported it to C for execution on a Blackfin DSP.
  - Co-authored the business plan and patent applications.
- 2009–2010 **Research Assistant**, *Jacobs University*, Bremen.  
PreCombine3D project: PMD ToF camera-driven 3D tracking. Implemented calibration and motion estimation software, as well as a camera Ethernet driver. Co-authored a research paper.

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## Skills

- Languages C++, C, GLSL, Swift, Objective-C, Haskell. Familiar with Go, Java, Python  
Expertise iOS Development, Unix/Linux, 3D Graphics, DSP, Computer Vision, L<sup>A</sup>T<sub>E</sub>X

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## Personal Projects

- Raytracer A CPU raytracer implemented in modern C++. Features include Blinn-Phong shading, shadows, bounding volume hierarchies, and path tracing.  
[github.com/cfr/raytracer](https://github.com/cfr/raytracer).

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## Master Thesis

- title *Simultaneous Localization and Mapping with a Time-of-Flight Camera*  
description The thesis presents a real-time visual navigation system using a time-of-flight camera without *a priori* knowledge of the scene. A method for fusing data from time-of-flight and optical cameras, as well as methods for increasing localization robustness by leveraging color and spatial information, are discussed.

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## Talks and Publications

- Arrows and Functional Reactive Programming, SPb Haskell User Group
- Introduction to Computer Vision, Chaos Constructions 2009
- A. Nuechter, S. Serebriakov, L. Stankevich: “Visual SLAM with a Time-of-Flight Camera”, Journal of Optical Technology, 2010